

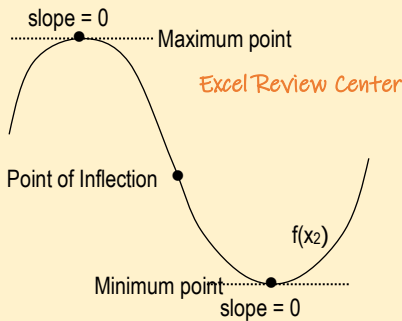


Solving Maxima/Minima Problems

The following are the steps in solving problems involving maxima and minima:

1. Draw a figure when necessary.
2. Determine which variable (the dependent variable) is to be maximized or minimized
3. Formulate equation
4. Reduce to one variable.
5. **Differentiate**
6. Equate to zero

When the first derivative (**slope**) is equated to zero, it results to either a maximum point or a minimum point.



	First Derivative	Second Derivative
Maximum Point	0	Negative
Minimum Point	0	Positive
Inflection Point	-	0

Solving Time Rates Problems

Another type of problems involving the use of derivatives are the time rates problems. In solving a problem under time rates, the following steps are to be considered:

1. Draw a figure when necessary.
2. Formulate equation.
3. Differentiate with respect to time.
4. Substitute the condition / instant to the equation.

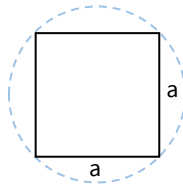
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Important: Substitute the given values only after differentiating.



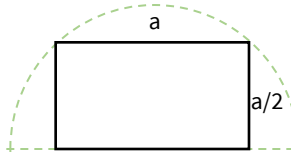
Variables in relation to Maxima/Minima Values

Largest Rectangle inscribed in a circle



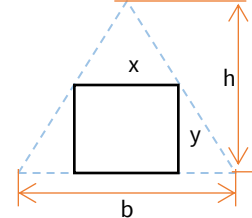
It is a SQUARE!

Largest Rectangle inscribed in a semicircle



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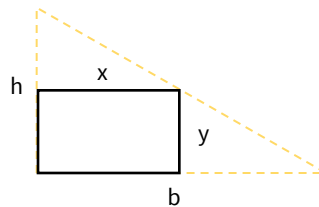
Largest Rectangle inscribed in a triangle with one side on the base



$$x = \frac{b}{2}$$

$$y = \frac{h}{2}$$

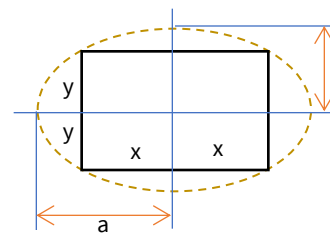
Largest Rectangle inscribed in a right triangle with sides parallel to the legs of the triangle



$$x = \frac{b}{2}$$

$$y = \frac{h}{2}$$

Largest Rectangle inscribed in an ellipse



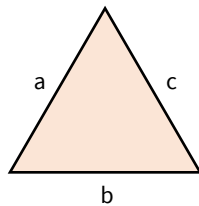
$$x = \frac{a}{\sqrt{2}}$$

$$y = \frac{b}{\sqrt{2}}$$

$$\frac{A_{\text{ellipse}}}{A_{\text{rectangle}}} = \frac{\pi}{2}$$

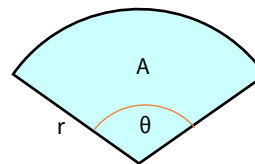
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Largest Area of a Triangle with given perimeter



$$a = b = c = \frac{P}{3}$$

Smallest perimeter of a sector with a given area

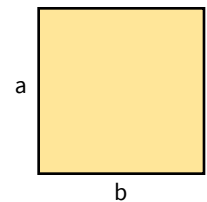


$$r = \sqrt{A}$$

$$\theta = 2 \text{ rad}$$

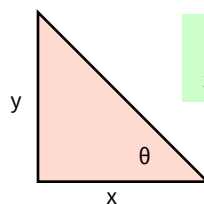
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Smallest perimeter of a rectangle with a given area



$$a = b = \frac{P}{4}$$

Right triangle with smallest perimeter or largest area

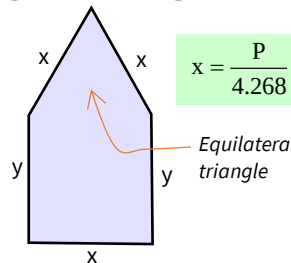


$$\theta = 45^\circ$$

$$x = y$$

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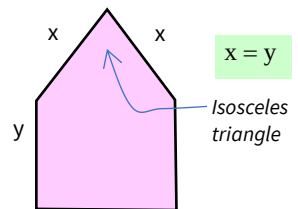
Largest area with perimeter given of the figure below



$$x = \frac{P}{4.268}$$

Equilateral triangle

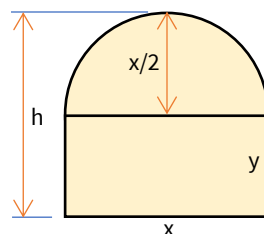
Maximum light admittance for the window shown



$$x = y$$

Isosceles triangle

Maximum light admittance for a Norman window

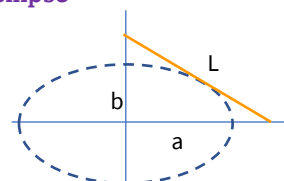


$$h = x$$

$$x = 2y$$

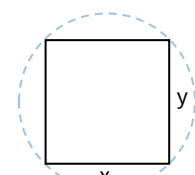
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Maximum length of line segment tangent to an ellipse



$$L = a + b$$

Rectangle of maximum perimeter inscribed in a circle



$$x = y$$

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